Goal: Build CRUD with PHP and MySQL

Learning Goals

- Define CRUD and match each operation to its corresponding SQL statement
- Explain the two-server architecture (PHP application server and MySQL database server)
- Explain why MySQL runs as an operating-system background service
- Verify MySQL installation, service status, and shell access

IMPORTANT

Our Goal Is CRUD

CRUD describes the four basic operations on stored data.

CRUD	SQL	What it does
Create	`INSERT`	Add a student
Read	`SELECT`	Find students
Update	`UPDATE`	Change a student
Delete	`DELETE`	Remove a student

PHP handles application logic. MySQL keeps the data. PDO connects them.

One Application, Two Servers

Last week:

```
Browser  →  PHP server
```

This week:

```
Browser  →  PHP server  →  MySQL server
                                  ↓
                           persistent data
```

- The PHP server runs our application code.
- The MySQL server stores and retrieves data for many requests.

How We Reach CRUD

```
1. Run MySQL
     ↓
2. Create the database, table, and PHP account
     ↓
3. Connect PHP to MySQL with PDO
     ↓
4. Send SQL from PHP to create, read, update, and delete data
```

This preparation lesson completes step 1. The next two core documents complete steps 2, 3, and 4 (followed by troubleshooting and security essentials in document 4).

IMPORTANT

Why MySQL Must Stay Running

MySQL is a separate server, not a library inside a PHP page.

- It stores data after a PHP request ends.
- It handles database files and SQL queries.
- PHP sends requests to MySQL and receives the results.

MySQL Runs as a Service

MySQL runs as an operating-system **service** — a background program kept alive continuously so it is always ready whenever PHP sends a query.

- Service management commands (`start`, `status`, `stop`, `restart`) are covered in: `course_tools/src/04. mysql installation.md`
- Re-read the "MySQL as a Service" slides there if you need a refresher.

Before You Continue: Confirm MySQL Is Ready

This document assumes MySQL is **already installed and running** —
that was HW1 / `course_tools/src/04. mysql installation.md`.

Quick check:

```
mysql --version
```

Ubuntu / WSL2:

```
sudo service mysql status
```

macOS:

```
brew services list
```

If either command shows MySQL is missing or not running, stop here and go back to `course_tools/pdf/04. mysql installation.pdf` (or `course_tools/code/04. mysql installation/run.sh`) and follow the install/service steps again before continuing.

Open the MySQL Shell

In your terminal:

WSL2/Ubuntu:

```
sudo mysql
```

macOS:

```
mysql -u root
```

Inside the MySQL shell, the prompt appears as `mysql>`. To exit back to your terminal, type `EXIT;`.

For password and shell-usage details, refer to `course_tools/pdf/04. mysql installation.pdf`.

Installation Checkpoint

Before continuing, confirm all three:

1. MySQL is installed.
2. The MySQL service is running.
3. You can open and exit the MySQL shell.

Next, we will create the persistent data used by our CRUD application.